TECHNO INDIA UNIVERSITY
WESTBENGAL

Stock Price Prediction and Management

8th Semester Project Report

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Techno India University, West Bengal

In partial fulfilment of the requirements
for the award of the degree of

Bachelor of Technology

in

Computer Science & Engineering

by

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DEDICATED TO

Firstly, to The Almighty God, whose blessings of grace, wisdom, and guidance have been the source of strength for us throughout this process.

To our esteemed Project Supervisor, Prof. Jayanta Chaudhuri, for his guidance and valuable advice, and to our Head of Department, Dr. Kalpita Dutta for her vision and leadership which created an enabling academic atmosphere for us.

To all the faculty members of the Department of Computer Science and Engineering, Techno India University, who have through their guidance in eight semesters laid down the strong foundation of knowledge for this work.

DECLARATION OF AUTHORSHIP

We hereby declare that the project report entitled **"Stock Price Prediction and Management"** is an authentic record of our own work carried out at the **Department of Computer Science and Engineering, Techno India University, West Bengal**, during the **8th semester of the academic year 2025–2026** under the supervision of **Jayanta Chaudhuri**, Assistant Professor.

We further declare that the matter embodied in this project report has not been submitted by us for the award of any other degree or diploma of this or any other Institute/University. All the information have been obtained and presented in accordance with academic rules and ethical conduct. We also declare that, as required by these rules and conduct, we have fully cited and referenced all materials and results that are not original to this work.

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CERTIFICATE OF RECOMMENDATION

This is to certify that the work embodied in this thesis entitled **"Stock Price Prediction and Management"** has been satisfactorily completed by **Mayukh Karmakar** (Roll No. 221001001204), **Tathagata Hui** (Roll No. 221001001201), **Suvojit Dutta** (Roll No. 221001001206), **Akshat Raj Verma** (Roll No. 221001001198), and **Debjyoti Sasmal** (Roll No. 221001001199). It is a bonafide piece of work carried out under my supervision and guidance at Techno India University, Kolkata, for partial fulfilment of the requirements for the awarding of the **Bachelor of Technology** in **Computer Science & Engineering** degree of the Department of Computer Science and Engineering, Techno India University, during the academic year 2025 - 2026.

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ABSTRACT

Retail investors and students often lack access to a single tool that connects equity forecasting, market context, and hands-on trading practice in one place. This project addresses that need by building a web-based Stock Price Prediction and Management platform that brings together machine-learning forecasting, news-driven sentiment analysis, and a risk-free simulated brokerage—all within a unified Flask and SQLite application.

At the core of the system is a dual-mode Linear Regression engine that projects closing prices seven days ahead. Users choose between a Close Price model, which draws on historical pricing patterns, and a Volume-Based model, which uses trading activity as its primary signal.

Each mode displays a catalyst badge alongside its RMSE score so users can evaluate forecast quality at a glance. Alongside the price projections, a multi-source news aggregator scores recent financial headlines as positive, negative, or neutral, giving users qualitative market context to complement the quantitative forecast.

The paper-trading module supports account registration, a virtual cash wallet, buy and sell order execution with adjustable commission rates, dividend logging, live portfolio valuation, and a searchable transaction history. Every completed trade automatically generates a downloadable plain-text invoice, while an in-app notification layer keeps users informed of order confirmations and account activity. Portfolio Snapshot reports can be produced on demand for any of five time windows—7, 30, 90, or 365 days, or the full account history—supporting structured performance review.

System reliability is maintained through size-based log rotation, HTTP middleware that records per-request latency and error counts, and a `/health` endpoint for external monitoring. Administrators access a dedicated dashboard exposing live traffic metrics, recent log entries, and a Dataset Management interface for uploading, inspecting, and removing the CSV files used to train the forecasting model. Taken together, these capabilities close the gap that typically exists between standalone prediction tools and a fully functional portfolio management environment, making the platform a practical resource for learning and experimentation in equity markets.

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Chapter 1

INTRODUCTION

Retail investors and students often juggle separate tools for price forecasting, news analysis, and portfolio tracking. This project combines all three in a single Flask application. The forecasting engine provides two modes: a Close-price model trained on historical data, and a Volume-based model that incorporates trading activity.

1.1 Background

Pattern-based price analysis and sentiment analysis each tell only part of the story—historical trends show where a stock has been, while news sentiment captures the market's current mood. This project merges both into a single decision-support layer.

1.2 Problem Definition

In the modern age, equity analysis often tends to be narrow-minded in its applications—machine learning models rarely come with trading interfaces, and it's not rare to find paper-trading applications missing predictive analytics. This project manages to resolve this by providing a playground that not only handles the current valid stock data but comes prepackaged with short-term price projection, performance metrics, news-based sentiment scoring, and a secure stock trading and brokerage system.

1.3 Project Overview

The Stock Price Prediction and Management platform is a self-contained web application that delivers:

- Short-term equity price forecasting using a Linear Regression model
- Automated sentiment scoring drawn from current financial news sources
- A paper-trading module with full portfolio oversight and an administrative control panel

1.4 Scope

1.4.1 In Scope

- End-to-end Flask web application
- Role-based access control (standard user and administrator tiers)
- Market data ingestion via yfinance (primary) and Alpha Vantage (fallback)
- Linear Regression price forecasting over a 7-day horizon, with Closing Price and Volume-Driven modes
- Multi-source news sentiment aggregation with fallback logic
- Portfolio management: holdings, buy/sell execution, dividend processing, virtual cash, and commission handling
- In-app alert system for trade events, billing statements, and analytical reports, with per-user preference controls
- Portfolio Snapshot reports across selectable periods (7, 30, 90, 365 days, or all-time)
- Automatic invoice generation (plain-text) for every Purchase, Sale, and Dividend event

- Admin dashboard covering brokerage configuration, transaction review, user management, company records, aggregate statistics, live HTTP metrics, system event logs, and dataset governance
- Operational monitoring via a `/health` endpoint tracking response times, error rates, and uptime
- Rotating structured logging to `logs/app.log`
- New database entities: UserNotification, NotificationPreference, ReportSnapshot, Invoice

1.4.2 Out of Scope

- Live brokerage connectivity or real-money order execution
- Advanced MLOps capabilities such as automated model retraining, version registries, or experiment tracking
- Enterprise-scale infrastructure including distributed caching, asynchronous task queues, or multi-tenant deployments

1.5 Technology Stack

Table 1.1: Technology Stack

Layer	**Core Technologies**	**Libraries**
Backend	Python, Flask	Flask-SQLAlchemy, requests, Aiohttp
Database	SQLite	SQLAlchemy (ORM)
ML & Analytics	Python	scikit-learn, pandas, numpy
Sentiment / NLP	Python	NLTK Vader, TextBlob, BeautifulSoup, newspaper3k, tena
Frontend	HTML/CSS, JS	Bootstrap, D3.js

Chapter 2

LITERATURE REVIEW

2.1 Stock Market Forecasting

The strategically proven choice for selecting Linear Regression over some of the other complex deep learning models is because Linear Regression is computationally cheap for short-term predictions and fast to run as documented by Lavanya and Gnanasskaran [1]. Lavanya and Gnanasskaran [1] also noted in their research that simple linear models were able to perform reliably, while keeping deployment overhead low.

2.2 Linear Regression for Stock Price Prediction

These studies suggest linear regression remains a practical choice for stock price prediction, especially when paired with well-chosen inputs. Sangeetha and Alfia [2] tested a linear regression method and found that it gives a good result without needing a lot of computing power, which is very useful for websites and apps where speed is important. We also found out that if you choose the right input data carefully, linear regression gives reliable predictions. Another big advantage of this method is that it is easy to understand as you can clearly see which factors are affecting the predicted price.

2.3 Multiple Linear Regression and Feature Selection

Many studies have shown that linear regression works well for short-term stock price prediction. Sangeetha and Alfia [2] found that it delivers reliable results without heavy computation, making it practical for web applications where speed matters. Their work also highlights that careful feature selection significantly improves accuracy. A further advantage is the model's interpretability—it's straightforward to see which factors drive the prediction.

2.4 Gaps Identified

Most of these studies use linear regression for price prediction, but few explore how predictions can feed into a portfolio management workflow. This project bridges that gap by combining regression-based prediction with engineered features, scraped news headlines for market context, and a full paper-trading system—complete with broker commissions, user trading, and an admin dashboard.

Chapter 3

OBJECTIVES

3.1 Project Objectives

- Price Forecasting: Generate 7-day ahead price projections using a Linear Regression model
- Sentiment Analysis: Retrieve and score recent news coverage to produce positive/negative/neutral polarity readings
- Decision Support: Combine directional price forecasts with sentiment scores into actionable trading signals
- Portfolio Simulation: Provide virtual cash management, position tracking, buy/sell workflows, brokerage fees, dividend processing, and a full transaction history
- Administration: Supply an admin monitoring panel and brokerage configuration controls

3.2 Scope Summary

This platform is made strictly for educational purposes; it is mainly used to track market sentiment and portfolio simulation.

Chapter 4

METHODOLOGY

4.1 System Requirements

4.1.1 Hardware Requirements (Minimum)

- Processor: Dual-core CPU or better
- RAM: 4 GB (8 GB recommended)
- Storage: 1 GB free space
- Internet: Required (market data + news sentiment)

4.1.2 Software Requirements

- Operating System: Windows / Linux / macOS
- Python: 3.7+
- Libraries: as listed in requirements.txt

4.2 Data Collection and Preprocessing

4.2.1 Market Data Acquisition

When someone types in a stock name, the system first checks if it already has a saved {TICKER}.csv file on the computer. If the file is already there and up to date, the app just uses it to save time. If the file is missing or too old, the app uses yfinance to download about two years of historical price data from the internet. If yfinance has an issue or fails to load, the system automatically switches to Alpha Vantage as a backup to get the data.

4.2.2 Data Cleaning

- Rows containing missing values are removed using dropna
- The Close price column serves as the primary predictive signal

4.3 Linear Regression Forecasting Method

The forecasting model is based on the previously done research [1, 2, 3], which works as follows:

- A forecast window of a week is selected to get an estimate of price movements.
- Now, based on this target labels are created by shifting Close columns by n number of days.
- The pattern and price of the stock are collected from the historical dataset.
- Now all properties are normalised by using a StandardScaler that is consistent [2].
- Then an 80/20 train-test split is applied before proceeding with the linear regression model.
- Therefore, the trained model forecast closing prices for the next 7 trading days.
- The accuracy of the model is given as RMSE based on the 20% test set.
- For observed systematic market tendencies a 4% calibration factor is implemented.

Hence, this design achieves a workable balance between the predictive quality and response speed that makes the model suitable for web applications [1, 4].

4.4 Sentiment Analysis Method

Multiple articles and headlines relating to the stock are retrieved from various news sources, on the basis of this a sentiment score is computed for each piece of content and then aggregated. The final output results in the overall polarity (+ve, –ve or neutral) along with a count of articles used in each category.

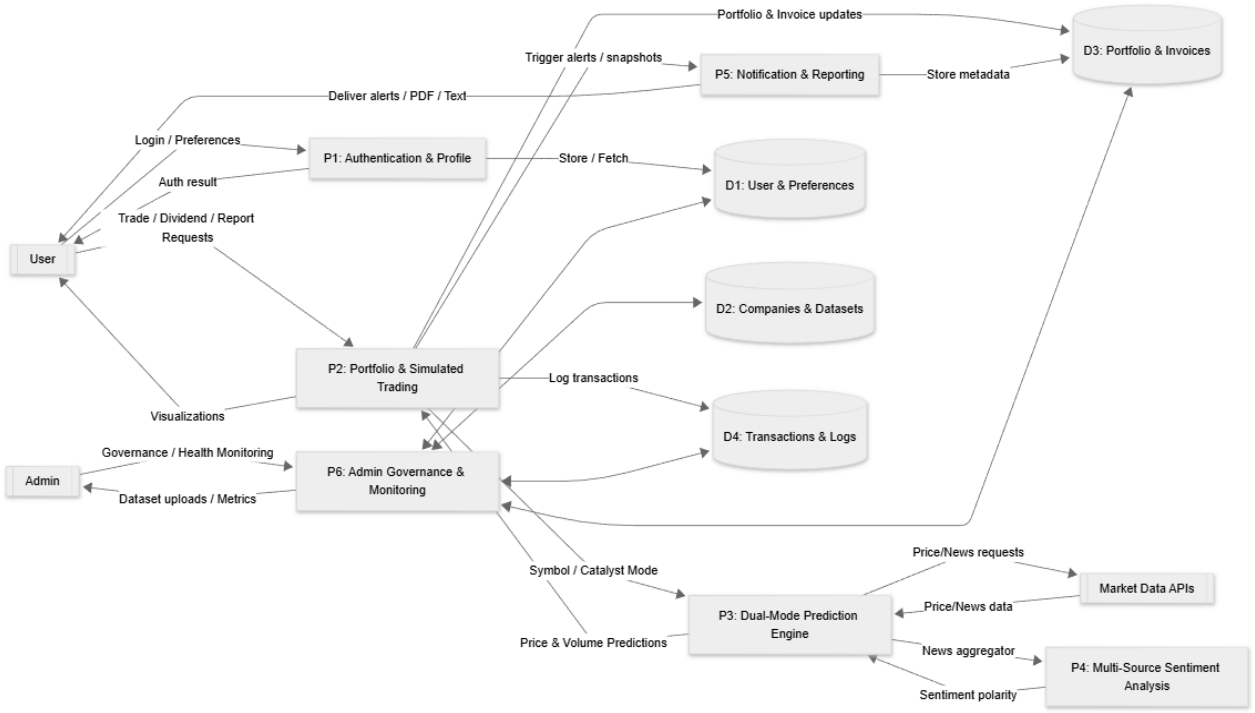
4.5 System Architecture and Design

4.5.1 Architectural Overview

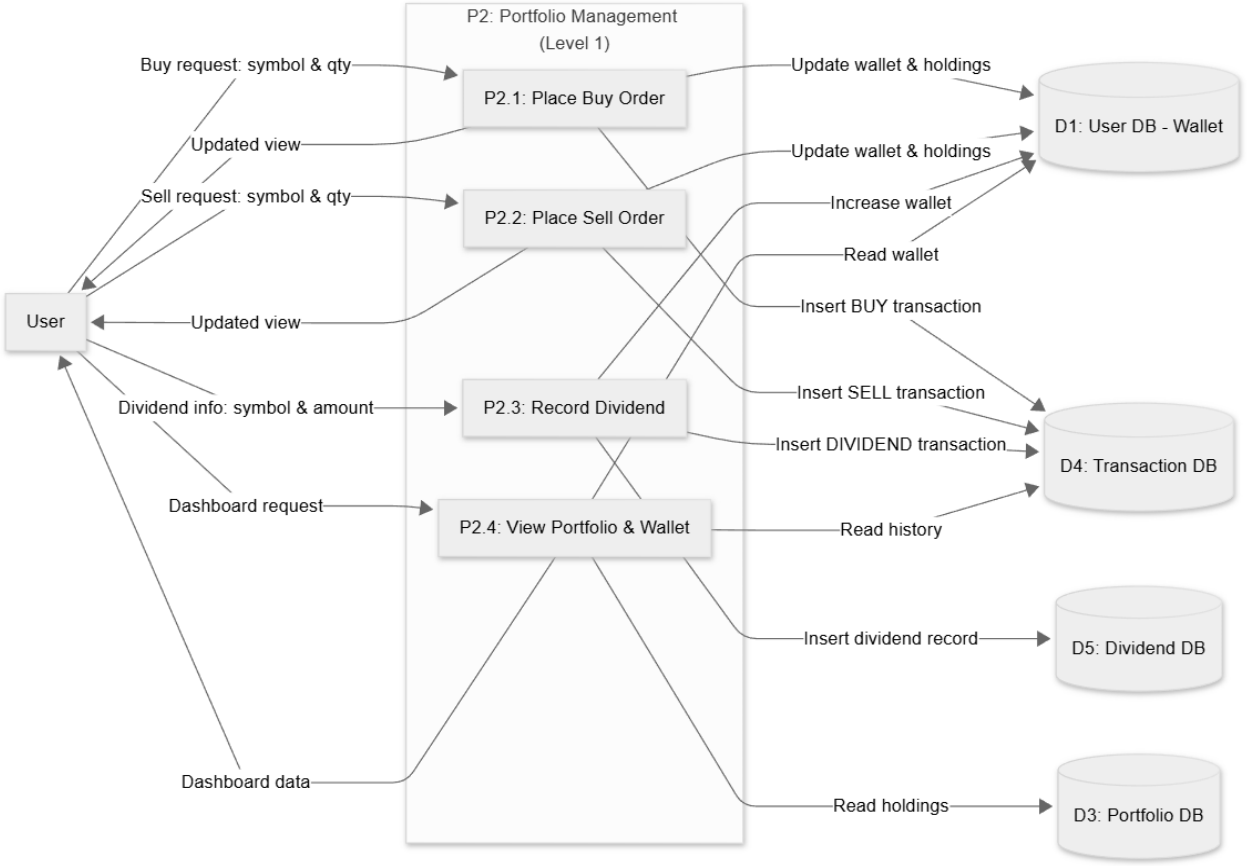
- Presentation Layer: HTML templates and dashboards
- Application Layer: Flask routes and business logic
- Data Layer: SQLite database via SQLAlchemy

4.5.2 Use Case Diagram (Figure 1)

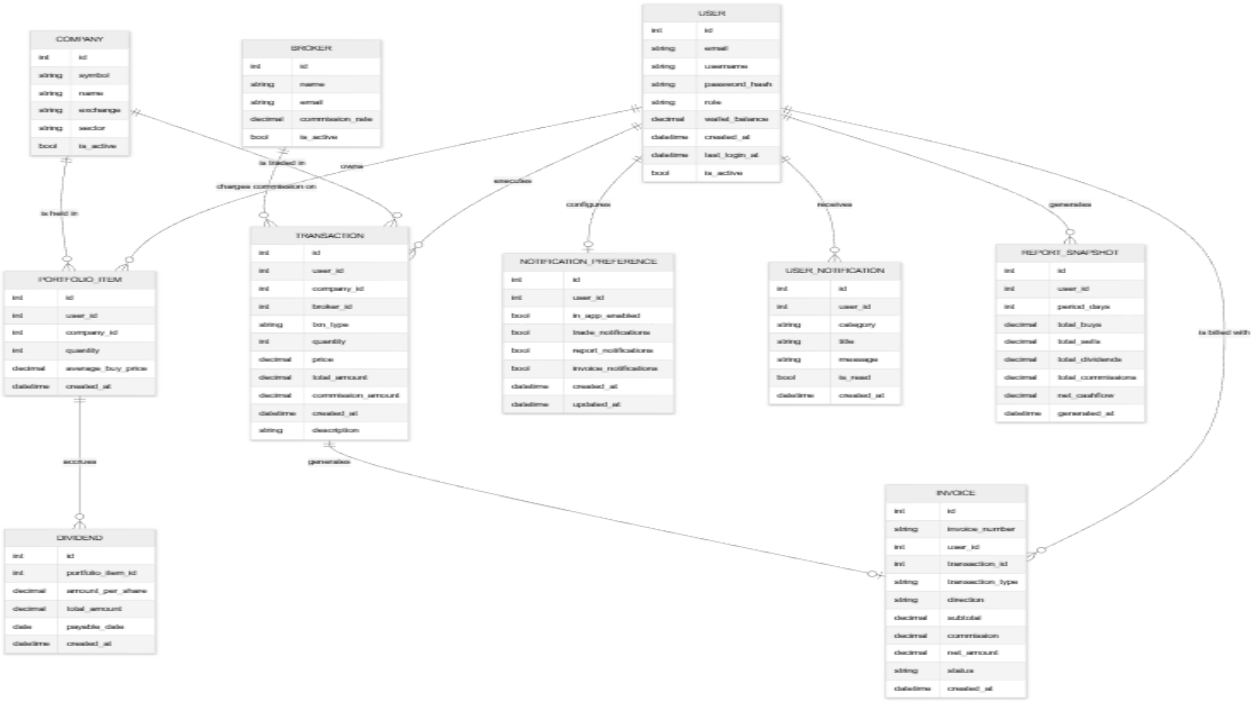
4.5.3 Data Flow Diagram (Level 0) (Figure 2)



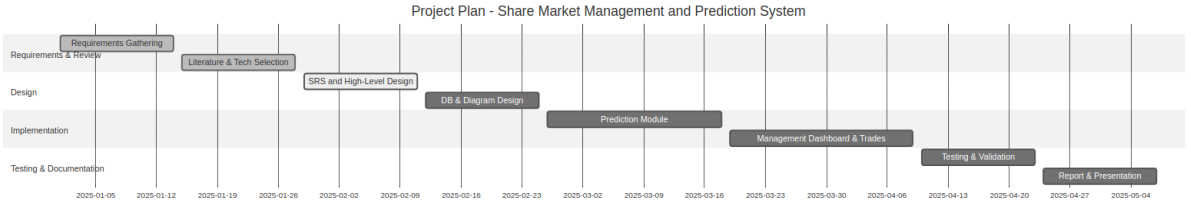
4.5.4 Portfolio DFD (Level 1) (Figure 3)



4.5.5 Database Design (ER Diagram) (Figure 4)



4.5.6 Project Plan (Gantt Chart) (Figure 5)



4.6 Implementation Overview

- **main.py:** This is the main file of the application. It handles all the Flask routes, defines the database models, and manages how the machine learning predictions are run.
- **news_sentiment.py:** A separate module that is used specifically for running the sentiment analysis.

- **The 'templates' folder:** A folder containing the Jinja2 HTML templates for the user dashboard, the prediction results page, and the admin panel.
- **static:** Contains all frontend assets: CSS, client-side JavaScript, and D3.js chart scripts.
- **Monitoring:** Tracks request metrics, exposes a health-check endpoint, and writes rotating log files for debugging.
- **Notifications & Settings:** Users can view alerts and update preferences in-app, backed by notification, report, and invoice tables.

Chapter 5

RESULTS AND DISCUSSIONS

5.1 Evaluation Metrics

Root Mean Squared Error (RMSE) is computed for the Linear Regression model and is displayed on the results page. This gives users an overview of the forecast accuracy against the test data.

5.2 Output Screens

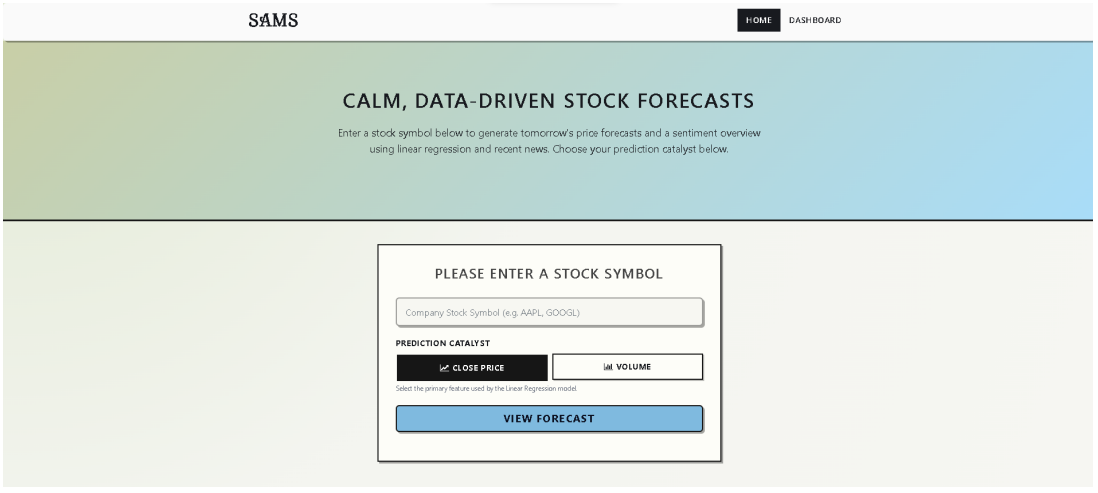


FIGURE 5.1: Home / Prediction Studio page

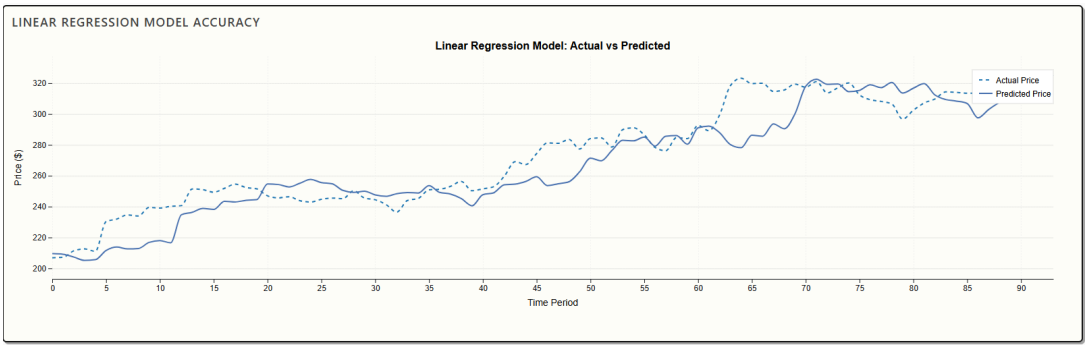


FIGURE 5.2: LR model test graph

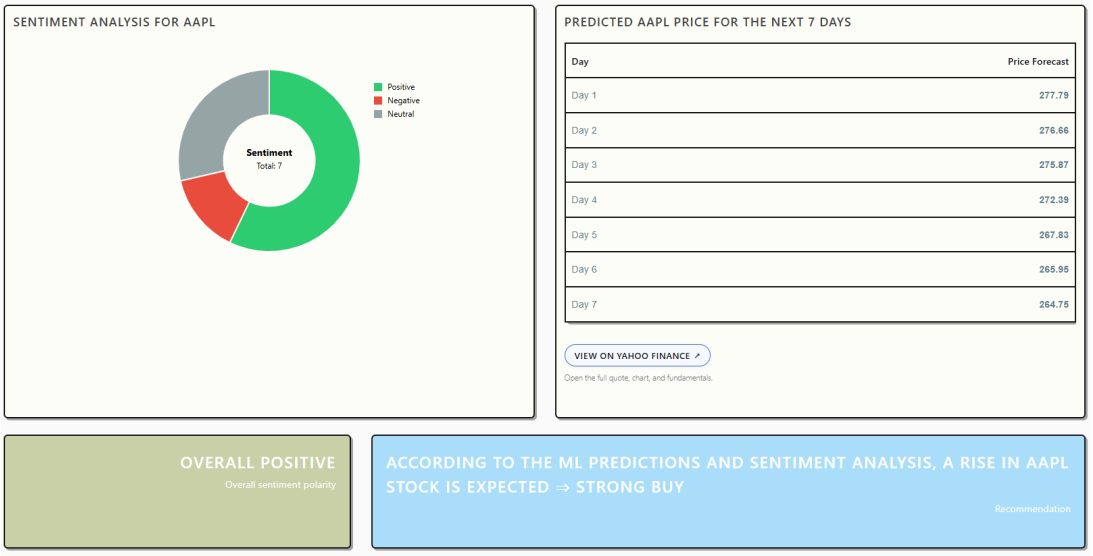


FIGURE 5.3: Sentiment Analysis

5.3 Discussion

- The system provides a unified view combining technical forecasting and sentiment context, addressing the integration gap identified in the literature [1], [5].
- The Linear Regression model delivers fast response times suitable for web applications while maintaining interpretable predictions [2], [3].
- Feature scaling and proper train-test splitting ensure robust model performance as recommended by recent studies [2], [4].
- The 7-day forecast horizon provides actionable short-term predictions that align with practical trading strategies.

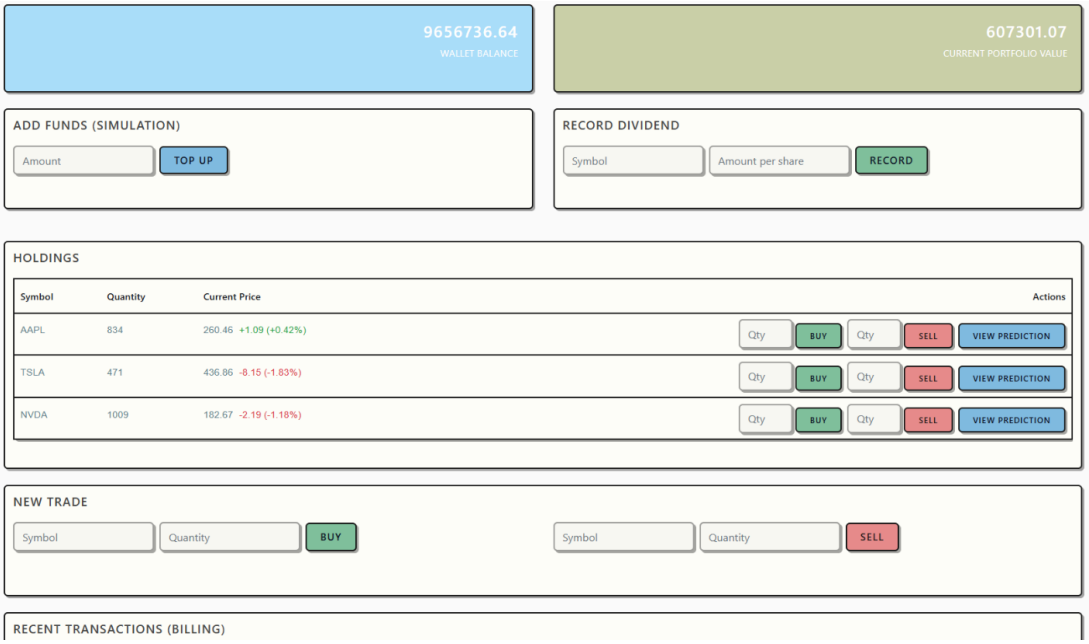


FIGURE 5.4: Stock Trading

FIGURE 5.5: Notifications

- The sentiment engine is designed with multiple fallback sources to remain operational even when primary data sources are unavailable.
- The combination of quantitative forecasting and qualitative sentiment analysis provides a more comprehensive decision-support framework than either approach alone.

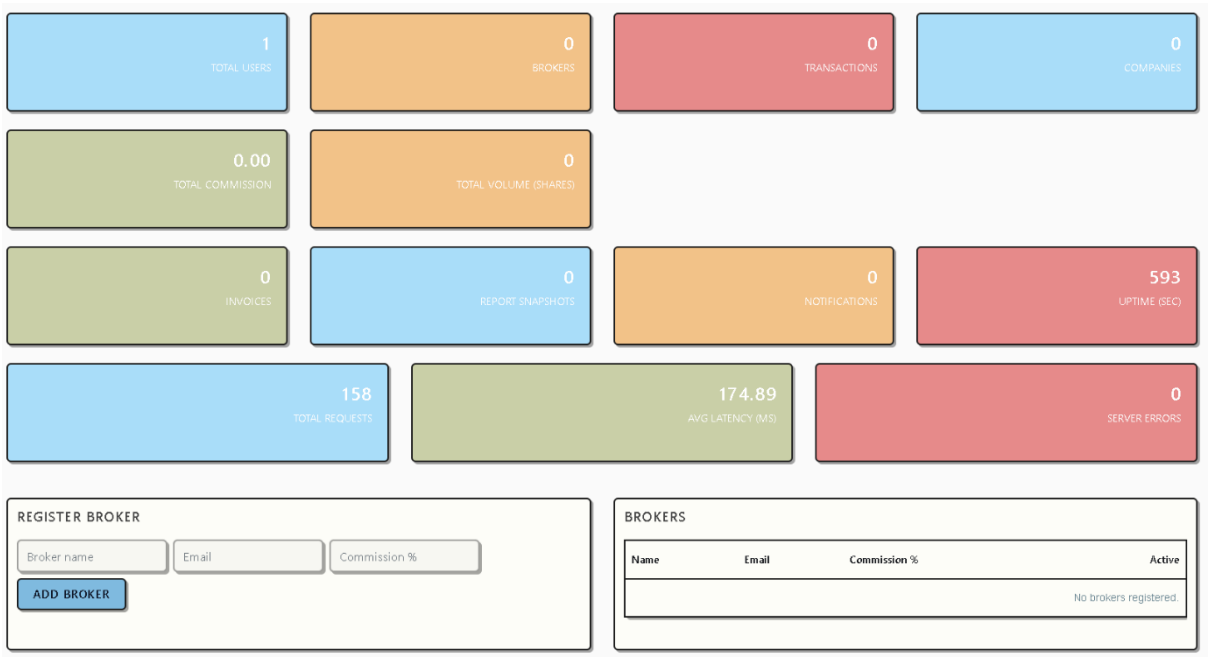


FIGURE 5.6: Admin Dashboard

FIGURE 5.7: Data Logging

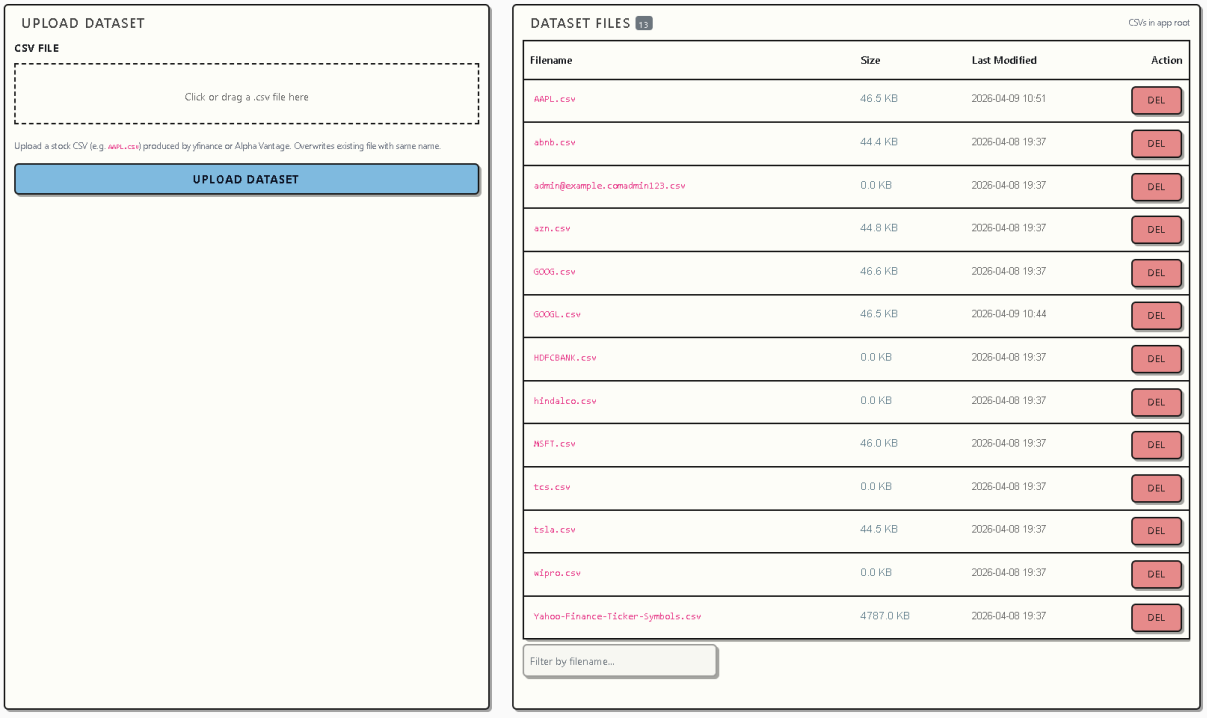


FIGURE 5.8: Dataset Management Interface

Chapter 6

FUTURE WORK

6.1 Conclusion

This system delivers a linear regression-based price prediction [1], [2], [3] combined with news sentiments aggregated from multiple sources and a robust portfolio management system. This system successfully proves that the heavy computation involved in deploying the website app remains steady all the while maintaining the effectiveness of the linear regression models short term prediction [4], [5]. This system serves as a means of preserving quantitative forecasting with qualitative sentiment analysis which proves this project has the potential of being a practical tool for learning and experimenting with stock market trading.

6.2 Future Enhancements

- Expanding the forecasting module to support additional models and richer feature sets.
- Strengthening validation through systematic testing (unit, integration, performance, and security testing).
- Full-scale Electron app deployment.

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[2] J. M. Sangeetha and K. J. Alfia, "Financial stock market forecast using evaluated linear regression based machine learning technique," *Meas.: Sensors*, vol. 31, p. 100950, Feb. 2024.

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Core Implementation Snippets

This appendix provides representative code snippets for the core analytical and operational modules of the Stock Price Prediction and Management platform.

.1 Linear Regression Forecasting Logic

The following Python snippet demonstrates the dual-mode forecasting engine using `scikit-learn`.

```
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split

def train_forecast_model(data, mode='close'):
    # Prepare features and labels
    if mode == 'volume':
        features = data[['Close', 'Volume', 'High', 'Low']]
    else:
        features = data[['Close', 'Open', 'High', 'Low']]

    # Shifting for 7-day forecast window
    labels = data['Close'].shift(-7).dropna()
    features = features.iloc[:-7]

    # Scaling
    scaler = StandardScaler()
    X = scaler.fit_transform(features)
```

```
    y = labels.values

    # Train/Test Split (80/20)
    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

    model = LinearRegression()
    model.fit(X_train, y_train)

    return model, scaler
```

.2 Sentiment Analysis Aggregator

The snippet below shows the logic for scoring headlines across multiple sources.

```
from textblob import TextBlob
from nltk.sentiment.vader import SentimentIntensityAnalyzer

def get_sentiment_score(headlines):
    sid = SentimentIntensityAnalyzer()
    scores = []

    for text in headlines:
        vader_score = sid.polarity_scores(text)['compound']
        blob_score = TextBlob(text).sentiment.polarity
        # Weighted average of multiple engines
        combined = (vader_score * 0.7) + (blob_score * 0.3)
        scores.append(combined)

    avg_score = sum(scores) / len(scores) if scores else 0
    return "Positive" if avg_score > 0.05 else "Negative" if avg_score < -0.05 else "
```

Database Data Dictionary

The platform utilizes a SQLite database managed via SQLAlchemy. This appendix details the primary entities and their attributes.

.3 Entity: Users

Field	Type	Description
id	Integer	Primary Key
username	String(50)	Unique username for login
email	String(120)	User contact email
password_hash	String(128)	Hashed password (Bcrypt)
role	String(20)	User or Admin tier
balance	Float	Virtual cash available for trading

Table 1: User Table Definition

.4 Entity: Trades

Field	Type	Description
id	Integer	Primary Key
user_id	Integer	Foreign Key to Users
ticker	String(10)	Stock symbol (e.g., AAPL)
trade_type	String(10)	Buy or Sell
quantity	Integer	Number of shares transacted
price	Float	Execution price per share
timestamp	DateTime	Date and time of the trade

Table 2: Trade History Table Definition

.5 New Operational Entities

- **UserNotification:** Stores in-app alerts for trade confirmations and system updates.
- **ReportSnapshot:** Stores JSON blobs of portfolio performance data for historical comparison.
- **Invoice:** Links to auto-generated plain-text billing records for every financial event.

Sample Artifacts and Reports

This appendix contains mockups of the automated outputs generated by the system for user record-keeping.

.6 Sample Plain-Text Invoice

Invoices are generated immediately upon trade execution or dividend processing.

```
========================================
      STOCK MANAGEMENT PLATFORM INVOICE
========================================
Invoice ID: INV-2026-0409-001
Date: 2026-04-09 14:30:22
User: Hare Krishna Chattopadhay
----------------------------------------
TRANSACTION DETAILS:
Type: BUY ORDER
Ticker: TSLA
Quantity: 10 Shares
Execution Price: Rs. 175.40
----------------------------------------
Subtotal: Rs. 1,754.00
Brokerage Fee (0.5%): Rs. 8.77
TOTAL DEDUCTION: Rs. 1,762.77
----------------------------------------
Status: COMPLETED
Thank you for using our simulated brokerage.
```

==========

.7 Portfolio Performance Snapshot

The following is a representation of the data captured in a 30-day snapshot.

```
{
  "snapshot_id": "SNAP-9921",
  "period": "30_DAYS",
  "start_balance": 10000.00,
  "end_balance": 10450.25,
  "net_return": "+4.50%",
  "top_performer": "NVDA",
  "total_trades": 12,
  "generated_at": "2026-04-10T09:00:00Z"
}
```